

LFCS Additional Operations Deployment Topics

Managing Kernel Boot Parameters



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Overview



Through this course:

- Kernel Options
- Red Hat Software
- SELinux

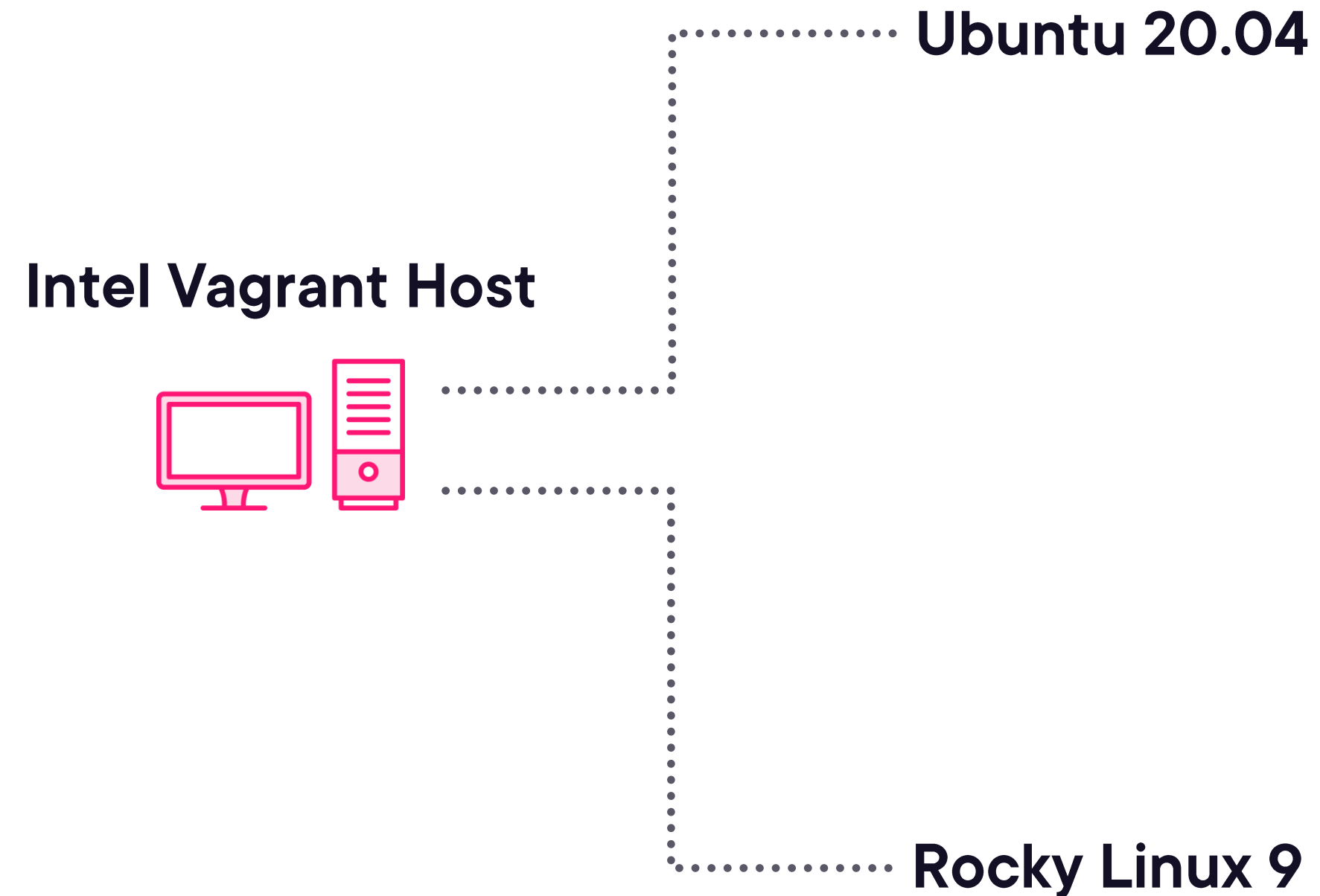
This Module

- Course Setup
- Grub Defaults
- Root Password Recovery
- Persisting Kernel Options



Lab Systems

<https://github.com/theurbanpenguin/lfcs>



Kernel options can be added transiently at the Grub menu or permanently added to the Grub configuration



```
$ sudo vim /etc/default/grub.d/99-custom.cfg
GRUB_TIMEOUT=20
GRUB_TIMEOUT_STYLE=menu
$ sudo update-grub
```

Ensure Grub Menu is Displayed

Ensuring the Grub menu is displayed would normally mean editing the `/etc/default/grub` file. However, on the Vagrant system there are additional configurations. You can ensure your configuration is read last by adding a file starting with `99` in the `/etc/default/grub.d/` directory. These are read in alphanumeric order so `99` is read last





GRUB Menu:

- To see the menu, you must use the hypervisor such as VirtualBox and not just Vagrant



Demo



Configuring the Grub Menu:

- Boot using VirtualBox
- Configure GRUB
- Reboot and view menu



```
e  
linux CTRL+E  
systemd.unit=rescue.target
```

Rescue Target

Booting to the rescue target in Ubuntu gives you root access without a password. You can set a root password or set a password for another account if it is unknown. This is not really a security risk as you need physical access to the server console to do this



Demo



Recover Password:

- Grub menu
- Add default target transiently



Recovery Mode:
Booting to recovery mode
can allow you to repair
filesystems and other
maintenance



Demo



Recovery Mode:

- Free space
- Manage broken packages
- Filesystem checks



```
$ sudo vim /etc/default/grub.d/99-custom.cfg
GRUB_CMDLINE_LINUX="quiet splash video=1366x768"
$ sudo update-grub
$ reboot
```

GRUB_CMDLINE_LINUX

To persist Kernel options, you can add them to the default GRUB configuration. As a demonstration you can set a larger 1366x768 screen resolution than the default. Set a lower resolution if your host is less than 1920x1024. The VM screen needs to fit on host screen



Demo



Persist Kernel Options:

- Set screen resolution for console



Summary



Reviewed the course contents:

- Kernel options
- Red Hat software
- SELinux

Kernel Options

- Configure Grub
- Set root password
- Working with recovery mode
- Persist Kernel options



Up Next:

Software Management on Red Hat Based Systems

